CANDIDATE NAME: ________________________________ INDEX NUMBER _________

CG _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H2 BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions

Tuesday
18 September 2018

Candidates answer on the Question Paper.
No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's use	
1	/27
2	/23
Section B	/25
Total	/75

This question paper consists of **21** printed pages including this cover page.

Section A

Answer **all** the questions in this section.

Question 1

(a) Outline the features found within a typical bacterium cell that distinguishes it from a typical plant cell.

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(b) One other way that bacteria differ from plants is in terms of how their genome is organised.

In bacteria, genes are generally organised into operons. Explain the advantage of this arrangement.

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(c) Like plant cells, bacterial cells are also surrounded by a rigid cell wall. However, while cellulose is a major component in plant cell walls, the significant constituent of the bacterial cell wall is peptidoglycan.

The peptidoglycan in some bacterial cell walls has the following features:

- It consists of linear chains of two alternating amino-sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM).
- In both NAG and NAM, the sugar component is covalently attached to a short oligopeptide sequence containing 4 to 5 amino acid residues.
- D-Alanine (D-Ala) and D-glutamine (D-Glu) are examples of amino acids that are commonly found in these sequences.

The structure of peptidoglycan in such bacterial cell walls is shown in Fig. 1.1 below.

A close-up of Region X is shown in Fig. 1.2 on page **4**.

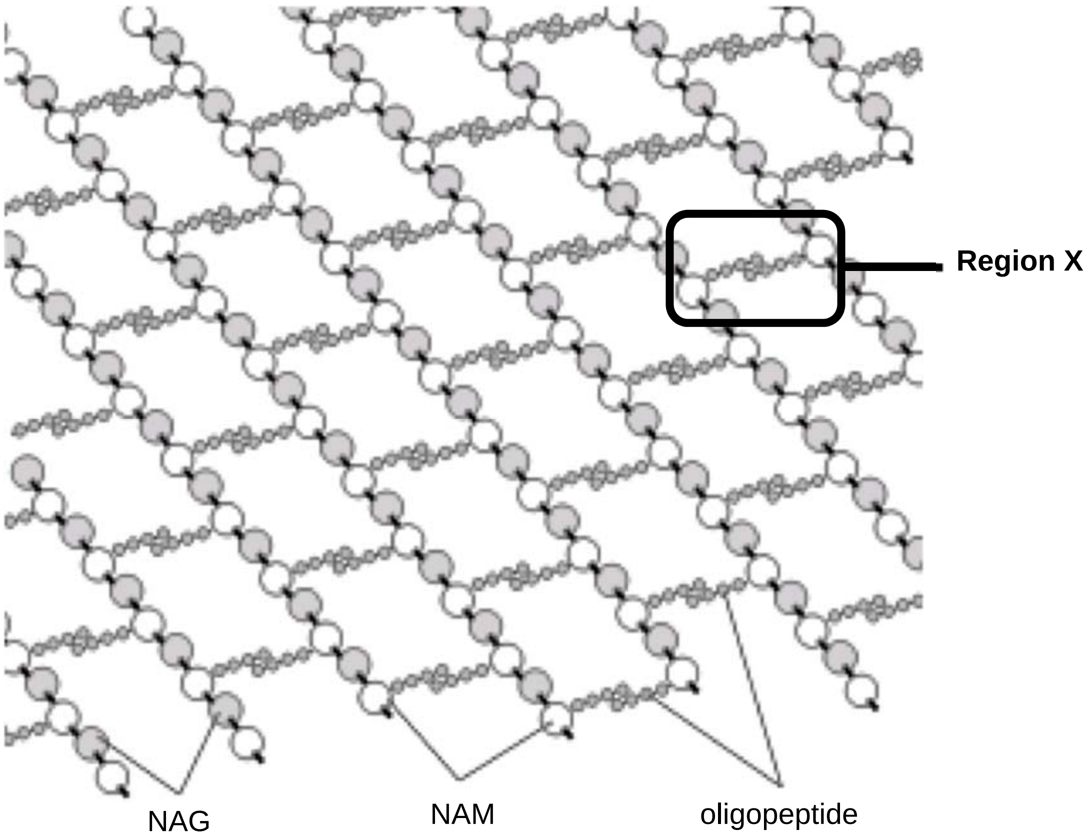
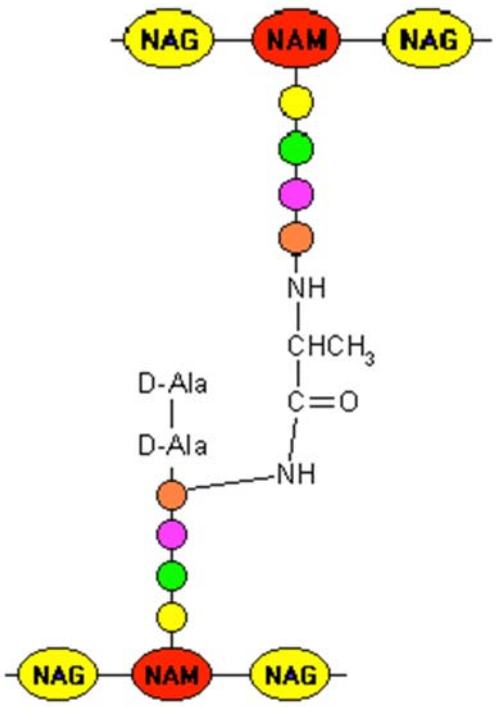


Fig. 1.1



Close-up of Region X

Fig. 1.2

(i) Describe how specific chemical groups of amino acids such as D-Ala and D-Glu allow them to be incorporated as part of an oligopeptide.

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(ii) Like cellulose, extensive cross-links are present in peptidoglycan.

With reference to the given information including Figs. 1.1 and 1.2, distinguish between the cross-links in cellulose and peptidoglycan.

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(iii) Suggest how the plant cell wall and bacterial cell wall are similar in function.

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(d) The formation of cross-links in peptidoglycan is catalysed by penicillin binding protein (PBP). A common antibiotic which targets this enzyme is penicillin. By inhibiting PBP, penicillin thus prevents the synthesis of the cell wall.

Fig. 1.3 below is a simplified sequence of events showing the action of PBP (**1** – **3**), and the action of penicillin on PBP (**4** – **5**).

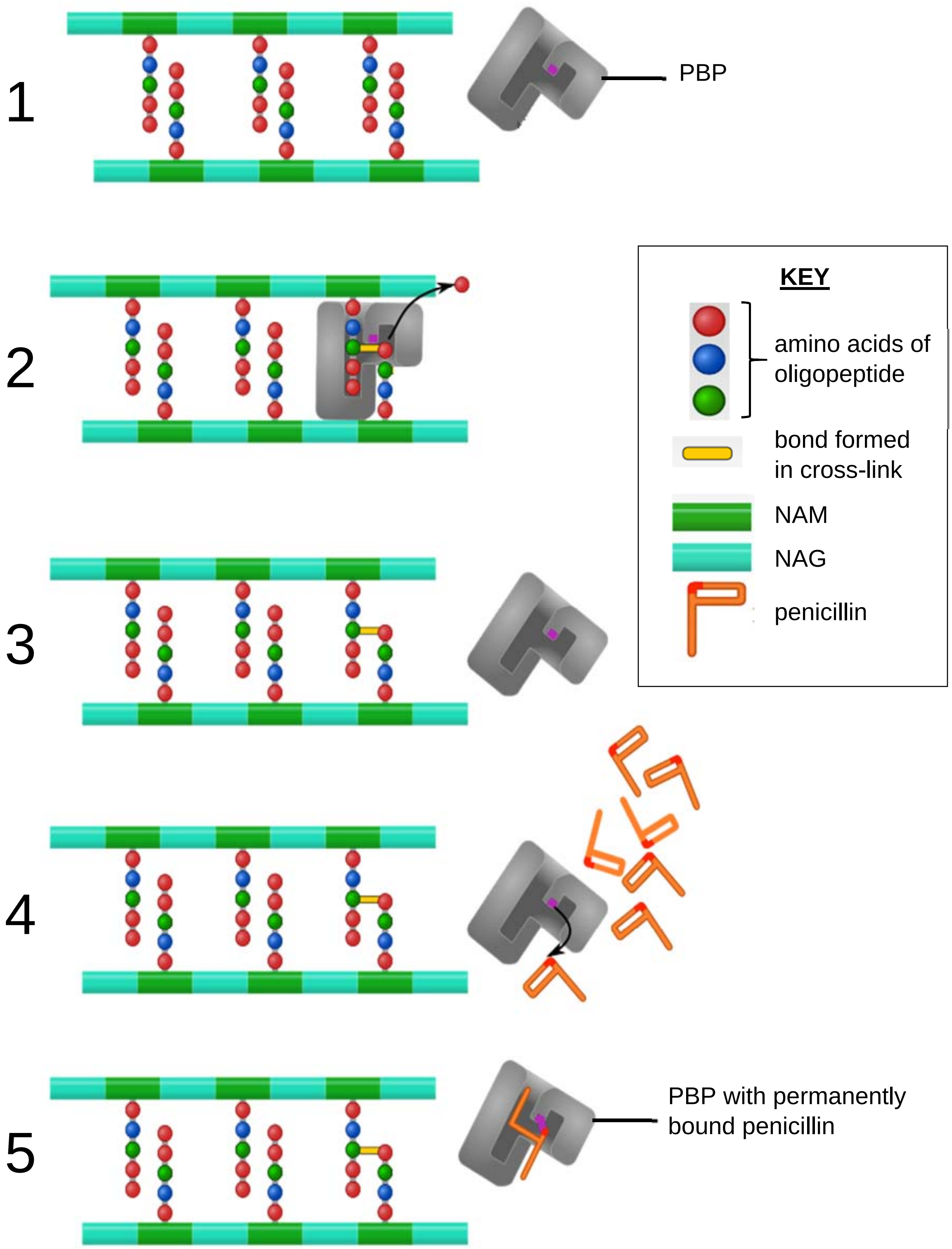

Fig. 1.3

(i) Although penicillin behaves like a typical competitive inhibitor, it is **not** a competitive inhibitor.

Compare the mode of action of penicillin with that of a typical competitive inhibitor.

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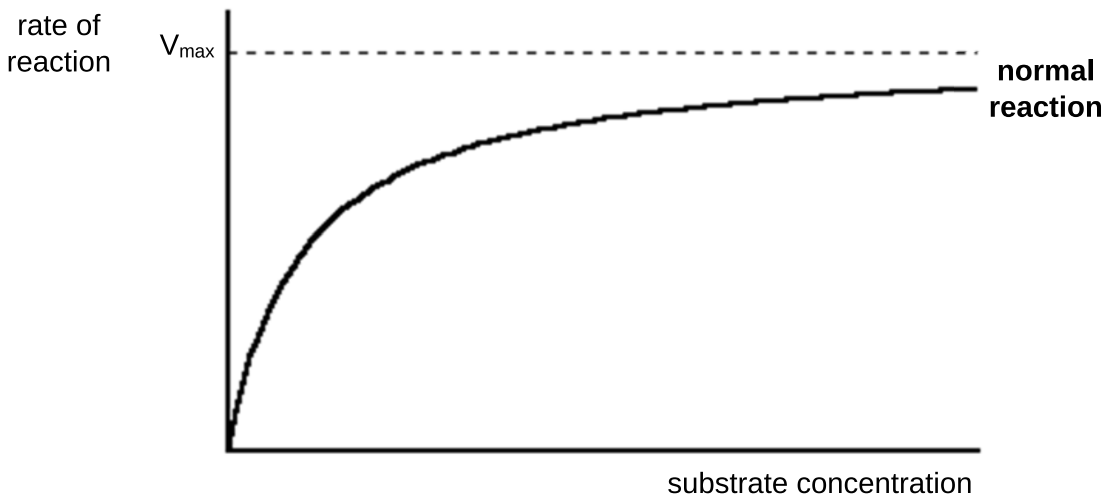
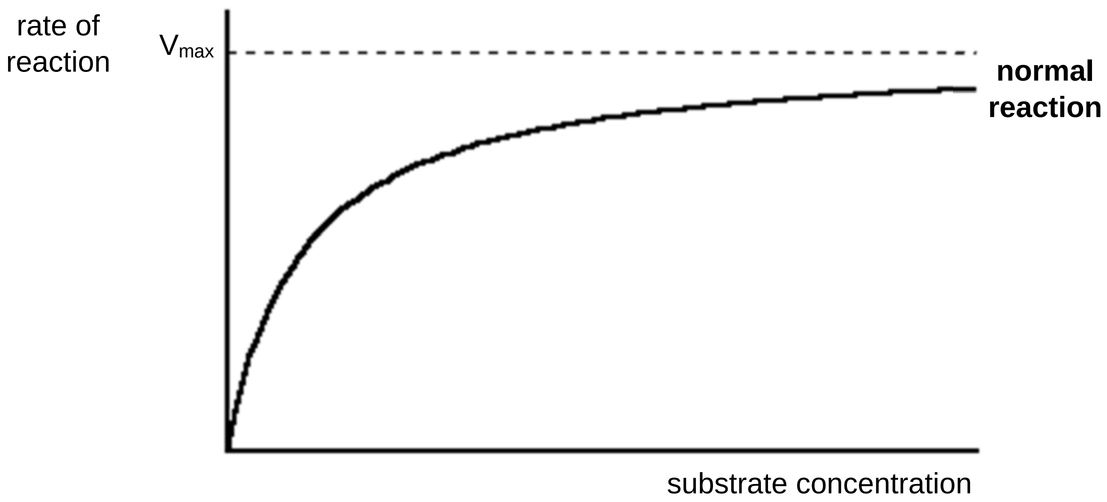
(ii) By drawing an appropriately labelled curve each, complete the graphs below to show the effect of

- a typical competitive inhibitor on its enzyme.



- penicillin on PBP.



[2]

(e) Another recently discovered antibiotic, SRJC30, inhibits bacterial growth by inhibiting protein synthesis.

SRJC30 functions by binding to the bacterial ribosome as shown in Fig. 1.4 below.

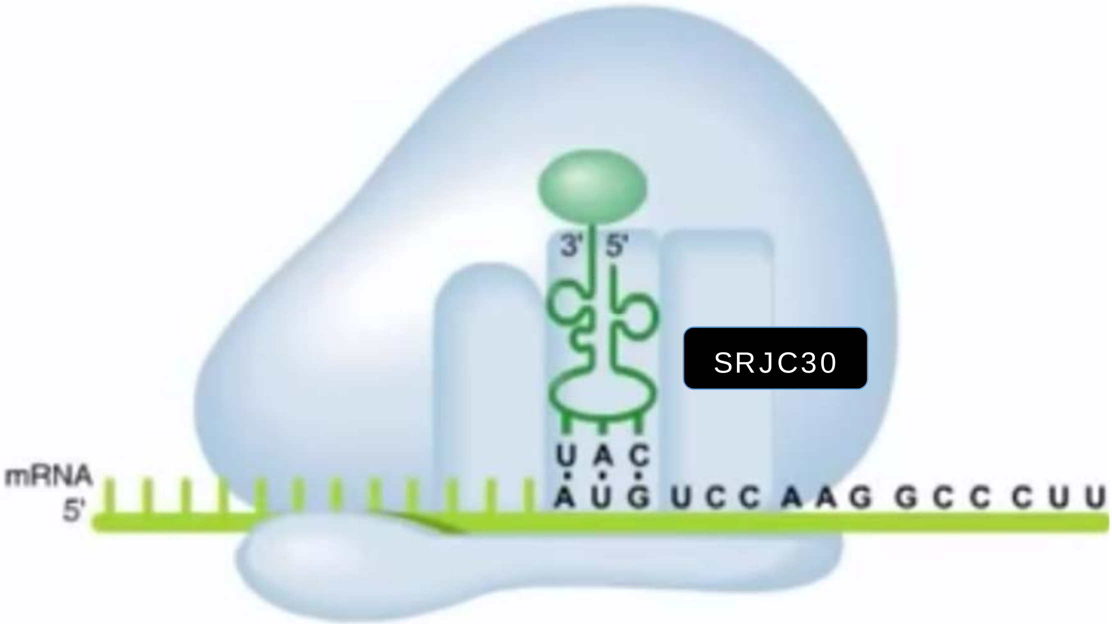


Fig. 1.4

Describe fully how SRJC30 inhibits protein synthesis.

In your answer, highlight **based on the context in Fig. 1.4**,

- the relevant specific locations on the ribosome and
- the specific molecules whose roles in protein synthesis have been prevented by SRJC30.

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(f) In a bacterial infection, the invading pathogen is recognised by the host's immune system as non-self. A co-ordinated and specific immune response is then mounted against it.

However, in treatment, the administered antibiotic is not recognised by the host's immune system, even when it is present in the bloodstream.

In modern medicine, antibiotics thus complement the host's immune system in fighting off a bacterial infection.

(i) Outline how a co-ordinated and specific immune response against a bacterial pathogen is possible.

In your answer, highlight the role of the specific immune cell that is directly involved in co-ordinating this response.

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(ii) Suggest why the host's immune system does not recognise an administered antibiotic.

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(g) Antibiotic resistance among bacteria is a growing global problem. Recent evidence indicates a relationship between antibiotic resistance and climate change.

Fig. 1.5 presents a finding from a study conducted in the United States, and shows a scatter plot of resistance among *E.coli* against the antibiotic amoxicillin, against minimum temperature.

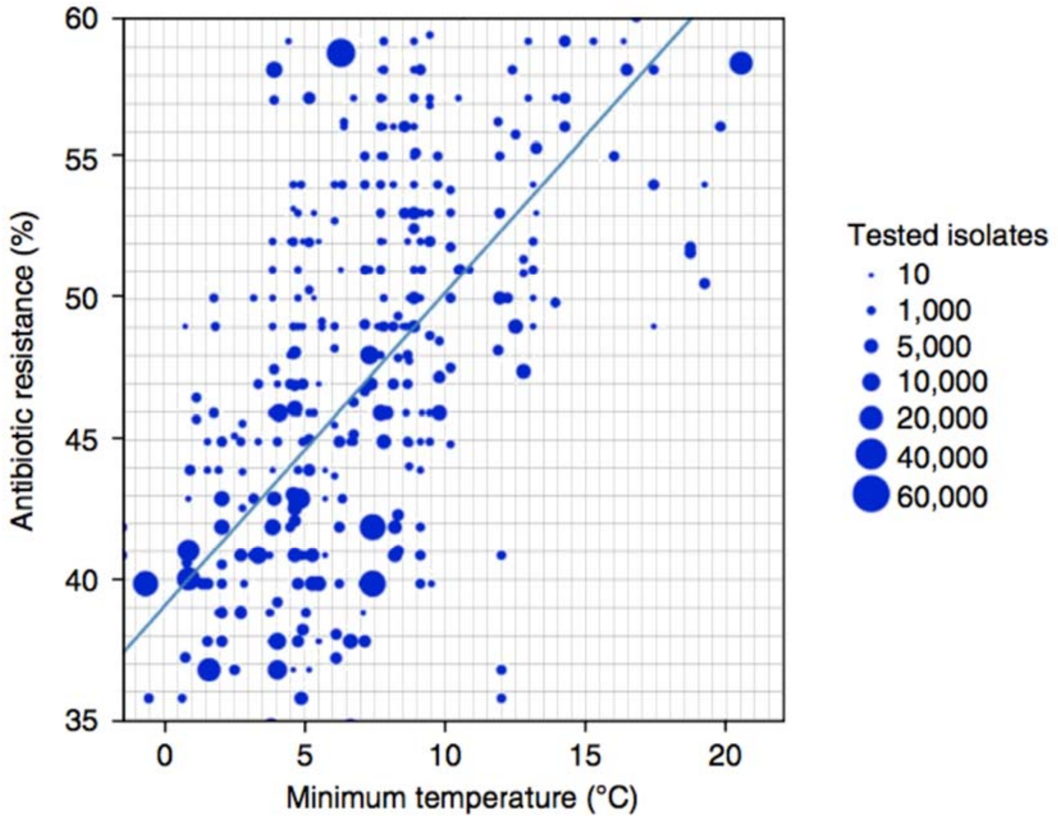

Fig. 1.5

Comment on how amoxicillin resistance among *E. coli* is affected with every 10 ^{o}C rise in minimum temperature, and suggest **two** reasons for this relationship.

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[Total: 27]

Question 2

Barley yellow dwarf virus (BYDV) is a positive sense single-stranded RNA virus; the viron is not enveloped in a lipid coating. The virus is transmitted by aphids, and the taxonomy of the virus is based on genome organisation, serotype differences and on the primary aphid vector of each isolate.

(a) Explain why viruses may be considered both living and non-living.

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(b) Using the information above and your knowledge, list two main classes of biomolecules present in a BYDV virion.

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(c) **(i)** Explain briefly how the virus is able to produce a complete new virion using the starting material of a positive sense single stranded RNA.

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(ii) Suggest one enzyme that is present in its host that the virus would need for the process explained in **(i)**, and one other enzyme that the host is unable to provide.

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The symptoms of a BYDV infection vary with the age of the plant at the time of infection, the strain of the virus and the environmental conditions. Symptoms appear approximately 14 days after infection. Affected plants may show a yellowing or reddening of leaves, stunting, an upright posture of thickened stiff leaves, reduced root growth, delayed (or no) heading, and a reduction in yield. Young plants are the most susceptible, and infected wheat leaves have a reduced ability to photosynthesise.

(d) Explain briefly how disrupting photosynthesis may lead to stunted growth.

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The host range of BYDVs consists of more than 150 grass species in the family *Poaceae*. A large number of grasses both annual and perennial are alternate hosts to the BYDV and can serve as reservoirs of the virus.

There are two main sources by which a cereal crop might be infected:

1. By non-migrant wingless aphids already present in the field and which colonise newly-emerging crops. This is known as "green-bridge transfer".
2. By winged aphids migrating into crops from elsewhere. These then reproduce and the offspring spread to neighbouring plants.

Transmission from an aphid is demonstrated in Fig. 2.1 below.

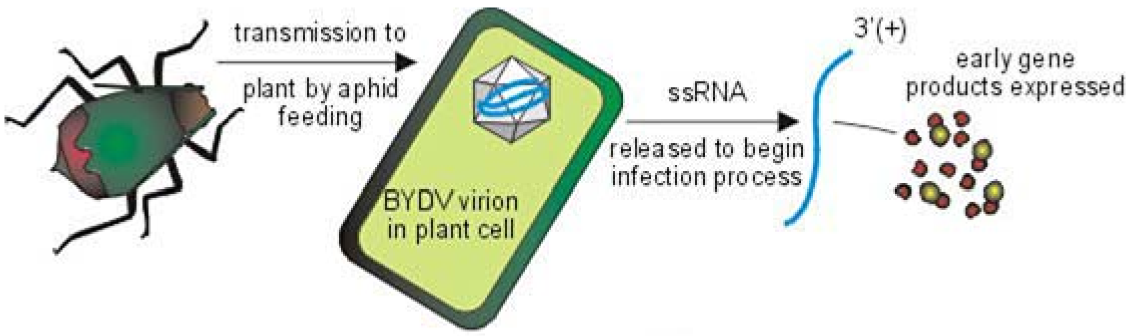

Fig. 2.1

(e) With reference to the grasses that serve as alternate hosts for the virus, explain what you understand about the concept of a species.

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(f) Climate change has affected the life cycle and distribution of animals and plants in the world.

Discuss how climate change may impact the rates of BYDV infection in a localised crop population.

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(g) Many farmers in rural farms in third world countries are struggling to cope with the impact of pests and pathogens such as the BYDV.

Suggest why many of these farmers are unable to overcome these issues.

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The following is an excerpt from a research paper on BYDV infections of two types of grass.

The wheat–Thin opyrum intermedium translocation line YW642 carries BYDV resistance gene BVDV-CP. To explore resistant wheat resistome in response to BYDV infection, we used GeneChip® Wheat Genome Arrays to analyze transcriptomes of YW642 and its susceptible parent Zhong8601 at 12 and 72 h postinoculation with BYDV.

Fig. 2.2 below shows the result of this investigation.

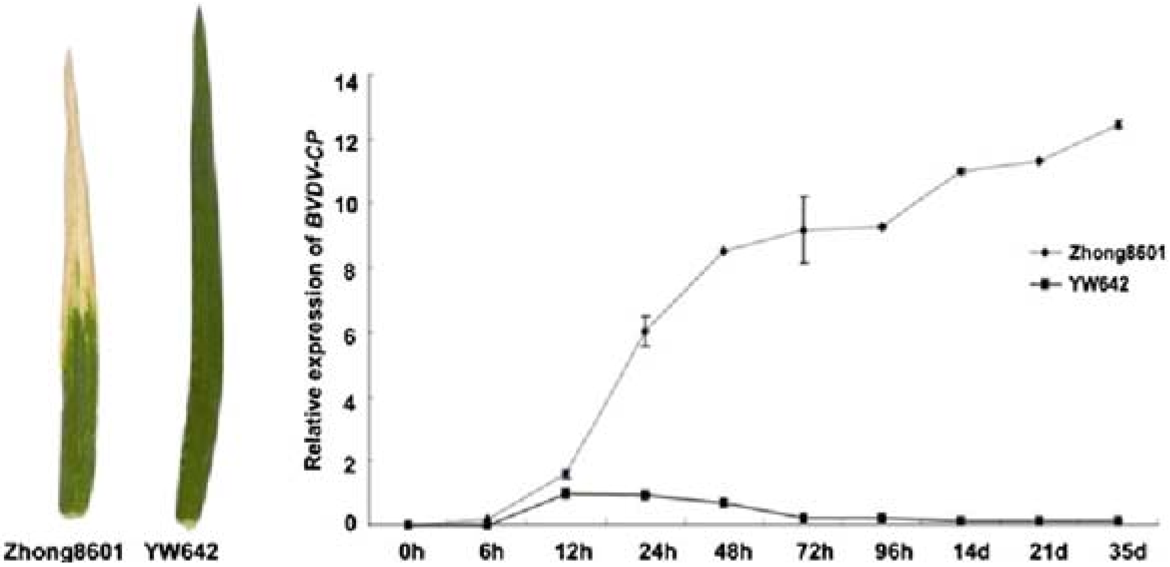

Fig. 2.2

(h) Using the information and data provided, describe the effects of inoculating the two grass strains with BYDV on the expression of the resistance gene *BVDV-CP*.

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(i) Suggest why the relative expression of the resistance gene *BVDV-CP* was low for both strains at 6 hours.

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(j) Suggest how the proteins coded for by *BVDV-CP* may lead to BYDV resistance.

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[Total: 23]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

Question 3

(a) Describe the principles and processes of the Polymerase Chain Reaction (PCR). [10]

(b) Discuss how photosynthesis plays a critical role in sustaining life on earth.

You should consider both specific processes in photosynthesis that sustain life and its role in reducing the impact of climate change. [15]

Question 4

(a) Describe the principles and processes of Southern Blotting. [10]

(b) It can be argued that humans have changed the levels of global respiration via economic activities and this has contributed to climate change.

Explain the processes in cellular respiration that release carbon dioxide and discuss how these and other human factors can contribute to climate change. [15]

[Total: 25]

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